Unified Health Index as a Service for Cross-Jurisdiction Emergency Care

Fernando LATORRE[a,c,1], Ivan BECERRO[b]. Nuria SALA[a,c].
[a] *Conéctate Soluciones y Aplicaciones, S.L., Soria, ES-CL, Spain.*
[b] *Accuro Technology S.L., Madrid, ES-MD, Spain.*
[c] *Fundación UNID, Soria, ES-CL, Spain.*

ORCiD ID: Fernando Latorre https://orcid.org/0009-0004-6313-6471
ORCiD ID: Nuria Sala https://orcid.org/0009-0001-0556-0694

Abstract

Data sharing spaces connect verified organizations under shared governance to enable cross-jurisdiction health information exchange without a central clinical repository. We present an implemented architecture for emergency location and continuity of care that uses decentralized identifiers (DIDs) and verifiable credentials (VCs) for provider discovery and role verification. Individuals are referenced by a Unified Health Identifier (UHI) implemented as an HTTPS-resolvable did:web identifier; resolving it returns a DID document with discovery metadata and service endpoints identifying the authorized provider hosting the individual's Unified Health Index (UHIx), while clinical records remain with original custodians (patient portals and HL7 FHIR endpoints). An individual, or a legal guardian acting under explicit signed consent, activates a UHIx at the chosen provider. This service externalizes a unified index of pointers and permissions, analogous to an international health booklet index, linking distributed records and storing fine-grained authorization rules. Policies are registered as auditable permit or deny rules under three scopes: professional role plus jurisdiction codes (country and optional region), professional role within a verified organization identified by tax ID plus optional department or office, or a specific verified email identity (e.g., a named clinician, caregiver, family member, or other related person). A blockchain network anchors discovery and policy artifacts for integrity, non-repudiation, and auditability via retrieval and hash and signature verification of as-registered artifacts. Verified responders authenticate and obtain access tokens with SMART on FHIR scope semantics to retrieve authorized International Patient Summary sections (ISO 27269 and HL7 FHIR IPS), enabling timely and traceable intervention.

Keywords

International Patient Summary (IPS), Patient Index (Unified Health Index), Cross-jurisdiction health information exchange, Decentralized identifiers (DID), SMART on FHIR authorization, Blockchain-based auditability.

1. Introduction

Data sharing spaces are emerging as a practical governance and infrastructure model to address cross-border challenges by connecting verified organizations under common

[1] Corresponding Author: Fernando Latorre Lopez; Email: ferlatorre78@gmail.com.

technical, legal, and operational rules (Gaia-X European Association for Data and Cloud AISBL, 2024).

Unlike centralized repositories, data spaces can enable trusted exchange while allowing clinical records to remain with their original custodians, such as patient portals, device manufacturers, and HL7 FHIR endpoints (HL7 International, 2019). In this setting, decentralized identity documents (DID documents) and W3C verifiable credentials (VCs) provide cryptographically verifiable attributes and roles used for authorization and auditing (e.g., verified provider, responder, gateway provider), which can be checked at runtime and traced to issuers under shared governance (World Wide Web Consortium, 2022, 2025). These mechanisms complement web PKI: HTTPS/TLS remains the standard for transport security and domain control, while VCs provide portable, multi-issuer attestations about entities and roles.

International Patient Summary (IPS) defines a minimal clinical dataset for unscheduled care and cross-border exchange (ISO 27269) and is commonly implemented using HL7 FHIR IPS profiles (International Organization for Standardization, 2025; HL7 International, 2024a). This paper presents an implemented approach centered on a Unified Health Index (UHIx) created at a gateway provider chosen by the individual.

The UHIx functions as a personal index of pointers and permissions, analogous to an international health booklet index, while the authoritative clinical data remains distributed across existing custodians. Access to distinct records exposed by patient portals and HL7 FHIR endpoints is obtained through standardized authorization and API access patterns such as SMART on FHIR (HL7 International, 2024b). To improve usability for people with disabilities or limited digital literacy, the design supports assisted enrollment and delegated administration, allowing a legal guardian to activate and manage an individual's UHIx under explicit, signed consent.

A key requirement for operational continuity is to separate stable identity anchors from changeable service locations. Our model uses a dual-identifier scheme: a stable, governed subject identifier (a semantic URN encoding network/jurisdiction/sector and a verified legal identifier), and a resolvable did:web identifier for the current connector domain used to discover public key material and service metadata through well-known HTTPS endpoints (W3C Credentials Community Group, 2024). This enables domain and infrastructure migration while preserving continuity of trust evidence, audit references, and authorization artifacts.

Emergency-grade interoperability requires fine-grained authorization that is both enforceable and auditable. Our implementation represents access rules as flat, interoperable claims aligned with HL7 FHIR search-parameter semantics, making the rule model independent of any particular downstream FHIR release used for rendering or exchange (HL7 International, 2019; Latorre, 2024).

In addition, a blockchain provides integrity and non-repudiation by recording provider discovery events and verifiable anchors of registered policy artifacts, enabling auditors to retrieve the as-registered artifacts and verify their integrity against immutable ledger records (European Commission, 2025). For European cross-border exchange guidance and Patient Summary alignment beyond the core IPS specification, we also consider the eHealth Network Patient Summary guideline (European Commission eHealth Network, 2024).

2. Method

2.1 Design approach and scope

We designed and implemented an end-to-end workflow for cross-jurisdiction emergency access to health information in a governed data sharing space. The method targets incidents where a verified responder must rapidly (i) identify the correct hosting provider for an individual's Unified Health Index (UHIx), (ii) obtain authorization limited to specific IPS sections and actions, and (iii) retrieve the corresponding records from distributed patient portals and HL7 FHIR endpoints (HL7 International, 2019; HL7 International, 2024a).

The workflow is built to function when the individual cannot actively interact with a device and to remain usable for people with disabilities or limited digital literacy through assisted enrollment and delegated administration by legal guardians or representatives.

2.2 Actors and verification

The method defines four actors:

- Individual (subject), owner of the UHIx and source of authorization.
- Legal guardian or representative, able to register and manage the subject's UHIx based on explicit signed consent.
- Verifiable organization, a member organization of the data sharing space operating service endpoints and issuing/verifying credentials under shared governance.
- Verified practitioner or responder, an employee of a verifiable organization requesting emergency access.

Organizations are onboarded through a shared-space trust process (e.g., verification by an intermediate credentialing authority) (Gaia-X European Association for Data and Cloud AISBL, 2024). Organizations register employee identities and roles and bind them to cryptographic authenticators generated by a confidential client application. Practitioners authenticate via an OpenID Provider (email-based identity) and prove possession of registered keys. All access requests are signed, and communications between users and services follow established high-assurance API security practices, including Financial-grade API (FAPI) security profile requirements for authentication and authorization flows (OpenID Foundation, 2021; OpenID Foundation, 2025).

2.3 Identity model and identifiers (stable anchor vs. resolvable locator)

We separate stable identity anchors from changeable service locations using a dual-identifier scheme:

- A stable, governed subject identifier (a semantic URN encoding network/jurisdiction/sector and a verified legal identifier) is used as the canonical reference for audit, deterministic identifiers, and continuity across operational changes.
- A resolvable did:web identifier represents the current connector domain used to discover public key material and service metadata via well-known HTTPS endpoints (W3C Credentials Community Group, 2024). This identifier may

change if the organization migrates domains or hosting infrastructure, while the anchored URN remains stable.

Verifiable credentials bind these elements by carrying the anchored subject identity (and its legal identifier claims) together with the resolvable identifier used for discovery and verification (World Wide Web Consortium, 2025).

2.4 Consent capture and enrollment options

UHIx enrollment can be completed in two equivalent ways:

- Self-service enrollment, where the individual creates the UHIx at a chosen provider using a signed consent.
- Assisted enrollment, where a member organization captures written consent (paper or digital), registers the service provider chosen by the individual, and creates the initial UHIx, access rules, and related persons as part of the same signed form. In jurisdictions with widely deployed qualified certificates, the consent can be implemented as a digitally signed PDF (e.g., using a national digital certificate).

In both cases, guardian authority and related persons are registered explicitly through signatures and can be audited (Latorre et al., 2023a, 2023b; Conéctate Soluciones y Aplicaciones SL, 2023).

2.5 IPS index externalization into UHIx

In HL7 FHIR, an IPS document is represented as a Composition with a structured index of sections and entries that reference underlying resources and linked documents. This index is typically assembled for a specific exchange and can remain coupled to the custodian that generates the IPS (HL7 International, 2019; HL7 International, 2024a).

The UHIx externalizes and unifies this index so it can exist independently of any single provider. Rather than storing clinical content, the UHIx stores pointers and permissions mapped to IPS sections, enabling section-level retrieval on demand under the authorization rules evaluated by the UHIx service. Records remain with original custodians (patient portals, device manufacturer portals, and HL7 FHIR endpoints), while the UHIx acts as the authoritative registry of where relevant items can be retrieved and under what conditions.

2.6 Discovery and blockchain responsibilities

The implementation uses the individual's hosted did:web identifier (UHI) for discovery and routing and the Unified Health Index (UHIx) as the personal index of pointers and permissions. DID resolution returns a DID document that can include verification material and service endpoints used for discovery and access workflows (World Wide Web Consortium, 2022). A did:web identifier is resolved by converting the method-specific identifier into an HTTPS URL and retrieving the DID document (did.json) at the location defined by the did:web method specification (W3C Credentials Community Group, 2024). The did:web method uses the web "well-known" location only when the identifier has no path segments, so that the DID document can be discovered under a fixed, root-level URL.

If the did:web identifier contains only a domain (no path segments), the DID document is retrieved from *https://<domain>/.well-known/did.json*; for example,

did:web:connector.example.com resolves to *https://connector.example.com/.well-known/did.json*. In case the identifier includes one or more path segments, the DID document is retrieved from *https://<domain>/<path>/did.json*; for example, *did:web:connector.example.com:individual:z3abc* resolves to *https://connector.example.com/individual/z3abc/did.json*.

The did:web specification standardizes the retrieval location of the DID document only. Auxiliary artifacts such as JWKS (JSON Web Key Set) documents, verifiable credentials, artifacts, and other metadata are exposed through HTTPS endpoints referenced from the DID document. This keeps discovery consistent while allowing the connector to define stable URL patterns for additional resources.

A blockchain is used for integrity and traceability of discovery and policy artifacts, not for clinical content. It anchors verifiable discovery and policy artifacts so that multiple verified inputs can resolve to the hosting provider of the individual's UHIx within the governed space. The anchored artifacts enable auditors to retrieve the as-registered documents and verify their integrity using hash and signature verification against immutable ledger records (European Commission, 2025).

2.7 Access rule model (three authorization scopes)

Access rules are expressed as signed policies that permit or deny access to IPS sections under one of three scopes:

- Jurisdiction and role: rules applying to practitioners by professional role within one or more jurisdictions.
- Organization and role: rules applying to practitioners by role within one or more verified organizations, including optional department or office constraints.
- Individual professional: rules applying to a specific professional identified by a verified email address, bound to a role and organizational credentials.

Each rule can include time bounds, section-level scope (IPS sections), and emergency constraints. Policy artifacts are anchored on the blockchain network for integrity and non-repudiation.

2.8 Authorization and SMART access token issuance

When a verified practitioner or responder requests access, the UHIx service evaluates the applicable rules and issues a granular SMART access token scoped to a specific endpoint action and IPS sections (HL7 International, 2024b). The token is then used to retrieve linked records from distributed portals and HL7 FHIR endpoints referenced by the UHIx, constrained by the approved scope. This enables practical enforcement at the API layer and limits access to minimum necessary data.

2.9 Linked identifiers, monitoring, and resilient secure exchange

In emergencies and catastrophes, additional identifiers (wearables, medical sensors, transport beacons) may be linked to the UHIx to support discovery, location, and time-bounded monitoring when authorized. Verified practitioners or responders request narrowly scoped access tokens to retrieve authorized IPS sections and follow UHIx links to distributed records (HL7 International, 2024b).

Beyond channel security, the system supports end-to-end protected exchanges using JWE/DIDComm-style encrypted payloads, enabling secure point-to-point communication without VPN dependencies and reducing exposure when intermediaries are untrusted or compromised (IETF, 2015; Decentralized Identity Foundation, 2022). This is particularly relevant to realistic threat models that include malware implants on endpoints or infrastructure components: message-level encryption and recipient-bound payload protection reduce the attack surface and limit the utility of intercepted traffic or stored artifacts. This does not eliminate endpoint compromise risk, but it reduces reliance on the confidentiality guarantees of the transport path alone.

2.10 Real-time device feeds and emergency telemetry

When the individual (or guardian) authorizes it, a wearable or medical sensor can publish real-time measurements referenced by the UHIx as a time-bounded stream endpoint (e.g., a FHIR Observation feed or a vendor API endpoint mapped to IPS-relevant sections). The UHIx stores the device identifier, the stream pointer, and the policy conditions under which a verified responder can request access. Upon authorization, the responder obtains a scoped SMART access token that enables read access to the specific streaming endpoint and the associated section(s) (e.g., vital signs), allowing near real-time monitoring during triage and transport (HL7 International, 2019; HL7 International, 2024b). This supports prioritization in mass-casualty incidents by providing objective signals (e.g., heart rate trends, oxygen saturation, temperature) under explicit consent-based rules and with auditable access.

2.11 Transport, disaster, and missing-person association

The method supports association of a UHI (or an authorized device identifier) to a transport asset (e.g., a vehicle device ID or beacon). If a crash or incident system can provide the associated identifier, emergency services can discover the relevant UHIx provider, request permitted IPS sections, and retrieve emergency contacts and critical clinical summaries without requiring manual identification (Latorre et al., 2023b; Conéctate Soluciones y Aplicaciones SL, 2023). A similar association applies to bracelets or tags for children, older adults, or missing persons, where an authorized responder can retrieve emergency contacts and limited emergency data defined by the individual or guardian.

2.12 Asynchronous API pattern and secure message model (JAR/JARM and message-level protection)

Except for well-known discovery and metadata endpoints (e.g., DID document retrieval, OpenID metadata, and other capability or key-discovery resources), the gateway models most state-changing operations as asynchronous jobs. A confidential software application submits a request, receives a polling URL, and later retrieves the final result. This pattern supports uniform auditing, rate control, and consistent handling of encrypted inputs and outputs, which is relevant in emergency operations where network conditions may be degraded or intermittent.

The security model aligns with Financial-grade API (FAPI) practices by using OAuth 2.0 request and response hardening mechanisms, including JWT Secured Authorization Request (JAR) and JWT Secured Authorization Response Mode (JARM), so that request parameters and responses are integrity-protected and bound to the intended client and server (OpenID Foundation, 2024a, 2025). In addition to transport

security (HTTPS/TLS), the implementation supports message-level protection using JWE-style encryption and, when applicable, DIDComm-style packaging, so that confidentiality and recipient binding do not rely exclusively on the communication channel (IETF, 2015; Decentralized Identity Foundation, 2022). The protected payload carries standard fields required for replay protection and correlation (e.g., issuer, audience, timestamps, nonce/jti, and a correlation identifier) together with the action-specific request body. This provides a consistent envelope for both index-maintenance and emergency workflows and reduces exposure when intermediaries, termination points, or network paths cannot be fully trusted.

2.13 Verified emergency online request under constrained communications

Beyond index maintenance and record retrieval, the implementation supports an emergency request initiated by the individual or legal guardian using their verified digital identity. This is relevant when voice channels are unavailable or saturated and when responders require an authenticated, traceable signal that a specific individual needs help. The emergency request is enveloped as a signed and encrypted message referencing the individual's UHI and the hosting provider of the UHIx (did:web), and it can include minimal, time-bounded emergency data and context (e.g., incident type, approximate location source, and optional information). The request is processed asynchronously and anchored for auditability under the same governance and consent model.

The individual is referenced using a did:web identifier that resolves through HTTPS to a DID document containing discovery metadata and service endpoints (World Wide Web Consortium, 2022; W3C Credentials Community Group, 2024). Message confidentiality and recipient binding are ensured through JWE-style encryption and, when used, DIDComm-style packaging, so that the emergency request remains protected even when transport paths or intermediaries cannot be fully trusted (IETF, 2015; Decentralized Identity Foundation, 2022)

2.14 Anchored research metadata and optional anonymization and digital twin generation

Beyond emergency access, cross-jurisdiction incidents such as epidemics and pandemics require rapid situational awareness based on aggregated signals. While clinical (private) content remains with custodians, the data space can support population-level analytics by allowing custodians to anchor minimal, non-identifying research metadata on the blockchain anchoring layer. Such metadata can include standardized clinical codes, event date or date bucket, jurisdiction (ISO country code and, where relevant, ISO region code), birth-year bucket, and administrative sex/gender as recorded by the custodian. Anchoring these structured attributes provides integrity and traceability for downstream statistics while limiting the exposure surface compared to anchoring full records. The anchoring design follows data minimization principles and avoids direct identifiers; when used, it supports time series analysis of diagnostic trends, evolution of tests, and cross-region comparisons in public health emergencies.

3. Discussion

This work shows that a governed data sharing space can support an emergency workflow that is interoperable, auditable, and operationally deployable without centralizing clinical data. The key design choice is to externalize the IPS "index" into a

Unified Health Index (UHIx) that stores pointers, discovery metadata, and fine-grained permissions, while clinical records remain with their original custodians (patient portals, device manufacturer portals, and HL7 FHIR endpoints). In practice, this reduces integration burden because the exchange surface can be constrained to IPS-aligned sections and standard retrieval mechanisms, while allowing responders to request only what is necessary (International Organization for Standardization, 2025; HL7 International, 2024a).

A second implication is that individuals (or legal guardians) can contribute user-generated information even when their custodians are not fully integrated into the space. In our approach, user-generated entries can be stored under the UHIx governance model and shared selectively when permitted. This supports offline-first operation: a personal application can maintain a ready-to-share IPS snapshot (or section subsets) and later deliver it securely to authorized practitioners, for example for second opinions or when connectivity is intermittent (Latorre et al., 2023a, 2023b).

3.1 Interoperability, resilience and IPS alignment

Using HL7 FHIR IPS profiles provides a pragmatic common denominator for cross-jurisdiction exchange, especially for unscheduled care. The UHIx model complements IPS by enabling section-level retrieval and assembly on demand: a responder requests specific IPS sections, and the software application retrieves only the authorized items from each custodian referenced in the index (HL7 International, 2024a; International Organization for Standardization, 2025). This pattern supports partial availability, where some custodians are reachable and others are not, which is common in disasters and cross-border incidents. Additionally, the individual can choose to store a copy of their data in one or more cloud services as backup copies to support emergency access when a custodian is unavailable (Conéctate Soluciones y Aplicaciones SL, 2023).

3.2 Security, privacy, and accountability

The approach combines DID-based discovery, signed consents, and a permissioned blockchain to provide traceability and non-repudiation for discovery and policy artifacts, while keeping clinical content off-chain. The blockchain anchors the individual's hosted did:web identifier and discovery hashes, together with tamper-evident traces of the public cryptographic material and authorized controllers (e.g., guardians or legal representatives) published in the individual's DID document, enabling auditable key lifecycle and delegated control (World Wide Web Consortium, 2022; W3C Credentials Community Group, 2024; European Commission, 2025).

Fine-grained access tokens scoped to specific actions and IPS sections provide a practical enforcement mechanism at the API layer and reduce overexposure. Authorization rules are represented as flat, interoperable claims aligned with HL7 FHIR search-parameter semantics, enabling consistent evaluation independent of downstream rendering into a particular FHIR release (HL7 International, 2019; Latorre, 2024).

A primary resilience requirement in emergency operations is functioning over untrusted networks and partially compromised infrastructure. For this reason, the system supports end-to-end protected exchanges using JWE/DIDComm-style encrypted payloads so that confidentiality and recipient binding do not rely exclusively on transport security or trusted intermediaries (IETF, 2015; Decentralized Identity

Foundation, 2022). This reduces the utility of intercepted traffic and limits exposure in scenarios where HTTPS may be terminated or degraded at intermediaries; endpoint compromise remains a separate threat that must be addressed through operational hardening and key isolation.

Building on the same message-level protection model, we model most state-changing operations as asynchronous jobs, providing a uniform enforcement point for authentication, audit logging, and encrypted request handling under degraded connectivity. The same mechanism can be used not only for retrieval requests but also for verified emergency requests initiated by the individual or guardian when conventional voice channels are unavailable or saturated, while maintaining traceability and non-repudiation under the same governance and consent framework.

3.3 Usability, accessibility, and delegated administration

Emergency access cannot depend on a person interacting with a device at the time of need. The method therefore supports assisted enrollment and delegated administration through legal guardians or representatives, while maintaining explicit signatures and accountability (Latorre et al., 2023a, 2023b), supporting inclusion of users with disabilities or limited digital literacy. In jurisdictions where qualified digital certificates are widely deployed, digitally signed PDFs. provide a realistic bridge between established consent practices and machine-verifiable authorization.

3.4 Governance and operational considerations

A data sharing space requires clear governance for organization verification, role lifecycle management, and incident response processes (Gaia-X European Association for Data and Cloud AISBL, 2024). Policy updates, revocation, and time-bounded emergency access must be operationally simple for individuals and guardians, while remaining enforceable for custodians. The blockchain provides a shared, tamper-evident anchoring layer for discovery and policy artifacts, but governance processes remain essential for onboarding, certification, and dispute handling.

3.5 Limitations and future work

This paper focuses on the implemented workflow and its security and interoperability rationale rather than large-scale performance metrics or clinical outcomes. Future work includes (i) empirical evaluation of time-to-discovery and time-to-first-section retrieval across heterogeneous custodians, (ii) systematic security testing of rule evaluation and token issuance, (iii) usability studies for assisted enrollment and guardian workflows, and (iv) extension of rule types (e.g., break-glass constraints, jurisdictional restrictions, and context-aware emergency policies), as well as broader integration of device telemetry and additional sectors.

Future work also includes evaluating emergency-request delivery latency and reliability across heterogeneous IP bearers (e.g., local resilient Wi-Fi deployments and other constrained networks) and assessing the privacy impact and utility of anchored, non-identifying research metadata for public health monitoring.

4. Conclusion

We presented an implemented, emergency-ready method for cross-jurisdiction health information access in a governed data sharing space. The approach uses a Unified Health Identifier (UHI), implemented as an HTTPS-resolvable did:web identifier, to

enable DID-based discovery of the provider hosting an individual's Unified Health Index (UHIx).

The UHIx is controlled by the individual or a legal guardian under explicit, signed consent, and it externalizes the International Patient Summary document index into section-level pointers and permissions. This enables on-demand retrieval of authorized clinical sections from distributed patient portals and HL7 FHIR endpoints while keeping clinical content with original custodians and off-chain.

A blockchain anchoring layer provides integrity, traceability, and non-repudiation for discovery and policy artifacts by anchoring discovery hashes and verifiable traces of relevant public cryptographic material and authorized controllers published in DID documents. Fine-grained access tokens scoped to specific actions and IPS sections enable least-privilege access and practical enforcement.

By supporting assisted enrollment, digitally signed consent forms, and delegated administration, the workflow remains usable for people with disabilities or limited digital literacy. Combined with end-to-end protected exchanges (JWE/DIDComm-style encrypted payloads) over untrusted networks, the method enables timely and auditable intervention during emergencies and catastrophes while preserving data minimization and custodian autonomy.

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